Ho Fung Tsoi

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Research Interests

LHC physics and building machine learning methods for accelerating scientific discoveries.
Experimental particle physics: new physics searches (compressed SUSY, exotic Higgs) and trigger system at the CERN LHC
Machine learning: self-supervised learning, foundation models, anomaly detection, symbolic regression
Fast inference on FPGAs: low-latency (sub-microsecond) machine learning algorithms on FPGAs

Education

Ph.D. Physics, University of Wisconsin–Madison 2024
Thesis: Search for Exotic Higgs Boson Decays with CMS and Fast Machine Learning Solutions for the LHC
Advisor: Sridhara Dasu

B.Sc. Physics, The Chinese University of Hong Kong 2018
Thesis: First Passage Time Problem of the Time-Dependent Ornstein-Uhlenbeck Process
Advisor: Chi-Fai Lo
Visiting at UC Berkeley & Berkeley Lab (Jan – Aug 2017)

Positions

Postdoctoral Researcher, University of Pennsylvania (Supervisor: Dylan Rankin) 2024 – Present

Awards

CERN, CMS PhD Thesis Award 2024
UWisconsin, Cornelius P. and Cynthia C. Browne Fellowship 2024
CUHK, CN Yang Scholarship 2017
CUHK, Wei Lun Foundation Exchange Scholarship 2017
CUHK, Professor Dennis Yam Kuen Lo Physics Award 2016

Selected Research Experience

Search for new physics at the LHC

- **Compressed SUSY.** ATLAS Run 3 search for electroweak production of supersymmetric particles in compressed mass spectra with low-momentum tracks (*2026–Present*).
- **Exotic Higgs.** CMS Run 2 search for exotic Higgs boson decays into pseudoscalars in the final state of two b quarks and two τ leptons, and the combination with the $\mu\mu$bb final state [4] (*2020–2024*).

Novel ML methods for high energy physics

- **Self-supervised learning / foundation models.** Self-distillation pre-training method **jBOT** for jets [12]; VICReg pre-training to improve resonance mass width reconstruction [13]; data-driven formation of invariant training pairs using energy mover's distance [14] (*2025–Present*).
- **SymbolFit.** Automatic discovery of parametric functions to model HEP data [6,11] (*2024–2026*).

Low-latency (sub-microsecond) ML algorithms on FPGAs

- **hls4ml.** Co-developer; an open-source tool that converts and optimizes trained neural networks into FPGA firmware [7] (*2023–Present*).
- **SpikePID.** Neuromorphic spiking neural networks (SNNs) for on-detector dN/dx clustering counting in future drift chambers [15] (*2025–Present*).
- **SparsePixels.** CNNs with sparse convolution for spatially sparse image data on FPGAs [8] (*2024–Present*).
- **SymbolNet.** Compression of neural networks using symbolic regression for fast inference [3,5] (*2022–2024*).

Trigger system at the LHC

- **GNTau.** Commissioning the GNN-based tau trigger at the ATLAS High-Level Trigger for Run 3 [9] (*2024–2025*).
- **CICADA.** Development of an anomaly detection trigger algorithm at the CMS Level-1 Trigger [10] (*2022–2024*).
- Development of the Data Quality Monitoring (DQM) software for the Calorimeter Layer-1 subsystem at the CMS Level-1 Trigger (*2020–2024*).
- HL-LHC trigger sensitivity projection for the CMS searches for exotic Higgs boson decays [1] (*2020–2021*).

Selected Publications

Listed below are selected publications to which I was a primary author (*: referred). I also co-authored papers by CMS and ATLAS as a collaboration member. Complete list at INSPIRE. Preprints at arXiv.

2026

16. SuperJEPA: Pretraining with a Supervision-Augmented Joint-Embedding Predictive Architecture for HEP — *Under review*
Dylan Rankin, **Ho Fung Tsoi** — arXiv:2610.36121

15. SpikePID: Real-Time Neuromorphic Cluster Counting in Future Drift Chambers — *Under review*
Ho Fung Tsoi, Kam Wai Lai, Dylan Rankin

*14. Similarity Pairing with Energy Mover's Distance for Self-Supervised Pre-Training at the LHC — *ML4PS 2026*
Ho Fung Tsoi, Dylan Rankin — arXiv:2609.17738

*13. Self-Supervised Learning for Robust Resonance Mass Regression in Cascade Decays — *ML4PS 2026*
Ho Fung Tsoi, Alex Yang, Luis Felipe Gutierrez Zagazeta, Shion Chen, Dylan Rankin — arXiv:2609.17726

*12. jBOT: Semantic Jet Representation Clustering Emerges from Self-Distillation — *SciPost Phys.*
Ho Fung Tsoi, Dylan Rankin — arXiv:2601.11719

11. Machine Can Automatically Discover Parametric Functions to Model HEP Data — *ICHEP 2026*
Ho Fung Tsoi, Dylan Rankin, Cecile Caillol, Miles Cranmer, Sridhara Dasu, Javier Duarte, Philip Harris, Elliot Lipeles — arXiv:2607.19750

10. Anomaly Detection in the CMS Level-1 Trigger in Run 3 — *Preparing for sub.*
CMS Collaboration — CMS PAS

9. Performance of the ATLAS Tau Trigger in Run 3 — *Preparing for sub.*
ATLAS Collaboration

2025

*8. SparsePixels: Efficient Convolution for Sparse Data on FPGAs — *Mach. Learn. Sci. Tech.*
Ho Fung Tsoi, Dylan Rankin, Vladimir Loncar, Philip Harris — arXiv:2512.06208

*7. hls4ml: A Flexible, Open-Source Platform for Deep Learning Acceleration on Reconfigurable Hardware — *ACM TRETS*
Jan-Frederik Schulte et al. — arXiv:2512.01463

2024

*6. SymbolFit: Automatic Parametric Modeling with Symbolic Regression — *Comput. Softw. Big Sci.*
Ho Fung Tsoi, Dylan Rankin, Cecile Caillol, Miles Cranmer, Sridhara Dasu, Javier Duarte, Philip Harris, Elliot Lipeles, Vladimir Loncar — arXiv:2411.09851

*5. SymbolNet: Neural Symbolic Regression with Adaptive Dynamic Pruning for Compression — *Mach. Learn. Sci. Tech.*
Ho Fung Tsoi, Vladimir Loncar, Sridhara Dasu, Philip Harris — arXiv:2401.09949

*4. Search for Exotic Decays of the Higgs Boson to a Pair of Pseudoscalars in the $\mu\mu$bb and $\tau\tau$bb Final States — *Eur. Phys. J. C*
CMS Collaboration — arXiv:2402.13358

2023

3. Symbolic Regression on FPGAs for Fast Machine Learning Inference — *CHEP 2023*
Ho Fung Tsoi, Adrian Alan Pol, Vladimir Loncar, Ekaterina Govorkova, Miles Cranmer, Sridhara Dasu, Peter Elmer, Philip Harris, Isobel Ojalvo, Maurizio Pierini — arXiv:2305.04099

2. Searches for exotic Higgs boson decays with the CMS experiment — *EPS-HEP 2023*
Ho Fung Tsoi *on behalf of the CMS Collaboration*

2021

1. The Phase-2 Upgrade of the CMS Data Acquisition and High Level Trigger — *CMS Technical Design Report*
CMS Collaboration

Conference, Workshop, Seminar Presentations

Fast Machine Learning for Science Conference (FastML) — UCSD, USA — Sep 2026
SpikePID: Neuromorphic Cluster Counting in Future Drift Chambers
SparsePixels++: Scalable Sparse Convolution on FPGAs

International Conference on High Energy Physics (ICHEP) — Natal, Brazil — Jul 2026
Machine Can Automatically Discover Parametric Functions to Model HEP Data

TREASURE: Tokenizing HEP Collider Data for AI Discovery — BNL, USA — Apr 2026
Self-Supervised Learning: Machine Can Learn a Lot by Just Observing

Coordinating Panel on Advanced Detectors (CPAD) — UPenn, USA — Oct 2025
SparsePixels: Efficient Convolution for Sparse Data on FPGAs

2024 CMS Thesis Award Ceremony Talk (CMS Week) — CERN, Switzerland — Oct 2025
Search for Exotic Higgs Boson Decays with CMS and Fast Machine Learning Solutions for the LHC

Fast Machine Learning for Science Conference (FastML) — ETH Zurich, Switzerland — Sep 2025
SparsePixels: Efficient Convolution for Sparse Data on FPGAs

High-energy physics seminar — UPenn/U.Washington/Fermilab, USA — 2023, 2024
Search for Exotic Higgs Boson Decays with CMS and Fast Machine Learning Solutions for the LHC

US LHC Users Association Meeting (US LUA) — Fermilab, USA — Dec 2023
CICADA: Anomaly Detection for New Physics Searches at the CMS Level-1 Trigger

Machine Learning at Level-1 Trigger Workshop (ML@L1) — CERN, Switzerland — Dec 2023
Anomaly Detection – CICADA: Status, Plans, and Prospects for Phase-2

CMS Machine Learning Town Hall — CERN, Switzerland — Sep 2023
L1 Anomaly Detection with Calorimeter Inputs: Status and Opportunities

European Physical Society Conference on High Energy Physics (EPS-HEP) — Hamburg, Germany — Aug 2023
Searches for exotic Higgs boson decays with the CMS experiment

International Conference on Computing in High Energy & Nuclear Physics (CHEP) — Norfolk VA, USA — May 2023
Symbolic Regression on FPGAs for Fast Machine Learning Inference

International Conference on Applied Mathematics — CityU, Hong Kong — May 2016
First Passage Time Problem of the Time-Dependent Ornstein-Uhlenbeck Process: a Model for Stochastic Decision-Making Process

Scientific Community Services

Membership

- Member, ATLAS Collaboration, CERN — 2024 – Present
- Affiliated member, Accelerated AI Algorithms for Data-Driven Discovery (A3D3) Institute — 2024 – Present
- Member, CMS Collaboration, CERN — 2020 – 2024

Collaboration Role

- **Co-convener**, SUSY Monte Carlo & Interpretation subgroup [CMS] — 2023 – 2024
- **Contact person**, Higgs boson Monte Carlo group [CMS] — 2021 – 2023
- **Software developer**, Data Quality Monitoring system for CaloLayer-1 trigger [CMS] — 2020 – 2024

Journal Referee

Machine Learning and the Physical Sciences (2026); *Transactions on Machine Learning Research* (2026); *IEEE Transactions on Evolutionary Computation* (2026); *Wiley Advanced Theory and Simulations* (2025).

Teaching and Mentoring

Teaching

- Guest lecturer [UPenn], Data Analysis for the Natural Sciences II: Machine Learning (PHYS 3359, Spring 2025)
- Teaching assistant [U.Wisconsin], undergraduate physics courses (Fall 2018 – Spring 2020)

Master students

- Kam Wai Lai [UPenn]: Spiking neural networks on FPGAs for cluster counting in drift chambers (2025–2026)

- Yuyan Wang [UPenn]: Joint-Embedding Predictive Architecture (JEPA) for jet anomaly detection (2025–2026)

Undergraduate students

- Chenjia Ni [UPenn]: Lorentz-equivariant neural network on FPGAs (2026–Present)
- Abhay Agarwal [UPenn]: Scalable sparse convolution on FPGAs (2026)
- Alex Yang [UPenn]: Mass regression of supersymmetric particles using contrastive learning (2025)
- Ashni Kumar [Drexel]: Acceleration of reinforcement learning agent inference on FPGAs for astronomical observations (2025)